Consistency. Let $\hat{\theta}_N$ be an estimator of θ based on a sample of size N. Say that $\hat{\theta}_N$ is a **consistent** estimator for θ if $\lim_{N\to\infty} \Pr\left(|\hat{\theta}_N - \theta| \geq \varepsilon\right) = 0$ for every $\varepsilon > 0$.

Consistency of ML Estimators. Suppose an ML estimator $\hat{\theta}$ of θ. Under certain *regularity conditions*, $\hat{\theta}$ is a consistent estimator of θ.